Box-Cox gamma sweep | Test ANN10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0342	0.0989	1.03	0.903	0.189	0.975	0.0714	0.988	1.01	0	18/18
1 ANN10	0.1	0.0321	0.142	1.26	1.01	0.17	0.99	0.0814	1.02	1.21	0	18/18
1 ANN10	0.2	0.0704	0.267	0.996	0.983	0.555	1	0.174	1.09	1.09	0	18/18
1 ANN10	0.3	0.0308	0.273	0.528	1.33	0.233	0.927	0.0761	2.59	5.57	0	18/18
1 ANN10	0.5	0.0607	0.52	0.763	0.94	0.507	0.808	0.0989	1.06	1	0	18/18
1 ANN10	1	0.0878	0.377	1.07	0.99	0.598	0.965	0.179	1	18.5	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Test P-only DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	0.051	0.188	0.946	1.56	0.243	1.42	0.164	0.935	0.999	0	18/18
2 P-only DeepONet10	0.1	0.0578	0.114	0.951	0.994	0.165	0.993	0.123	0.995	1.19	0	18/18
2 P-only DeepONet10	0.2	0.0727	0.368	1	0.961	0.202	0.857	0.208	0.987	1.01	0	18/18
2 P-only DeepONet10	0.3	0.168	0.575	1	1	0.533	2.17	0.516	0.99	0.999	0	18/18
2 P-only DeepONet10	0.5	0.342	0.769	3	1.57	0.773	1.48	0.498	0.997	13.3	0	18/18
2 P-only DeepONet10	1	0.513	0.518	1.34	1.97	1.17	1.02	0.919	1.06	21.5	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Test ANN10 + P-only DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	0.0509	0.17	0.989	0.938	0.263	0.994	0.128	0.975	0.999	0	18/18
3 ANN10 + P-only DeepONet10	0.1	0.0373	0.0777	0.901	0.92	0.244	0.993	0.0941	0.995	1	0	18/18
3 ANN10 + P-only DeepONet10	0.2	0.0352	0.263	1	0.677	0.17	0.629	0.0924	0.975	0.998	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.0502	0.347	0.976	0.953	0.266	0.833	0.158	0.986	0.998	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.128	0.513	1.25	1.26	0.602	1.6	0.192	0.915	14.2	0	18/18
3 ANN10 + P-only DeepONet10	1	0.116	0.427	0.939	0.98	0.66	0.934	0.52	1.08	16.8	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Test T/P DeepONet10: common ICs [5, 7, 18, 39, 44, 45, 53, 57, 64, 68, 69, 73, 76, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	0.21	0.394	1	1.08	0.672	1.03	0.315	2.12	1.83	0	18/18
4 T/P DeepONet10	0.1	0.0761	0.19	1.11	0.779	0.358	1.01	0.141	1.1	1.18	0	17/18
4 T/P DeepONet10	0.2	0.0184	0.1	0.621	0.79	0.379	0.619	0.205	0.996	1	0	18/18
4 T/P DeepONet10	0.3	0.426	0.904	0.978	14	1.02	1.65	0.574	3.92	1.73	0	18/18
4 T/P DeepONet10	0.5	0.326	0.632	1.84	1.33	0.697	1.02	0.463	13.4	7.18	0	16/18
4 T/P DeepONet10	1	0.432	1.22	1.18	3.28	0.902	0.671	0.71	1.01	40.5	0	17/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Test ANN10 + T/P DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	0.0747	0.616	1	0.962	0.563	1.03	0.18	1.5	2.4	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.0491	0.0957	0.962	0.82	0.174	0.97	0.066	0.921	0.993	0	18/18
5 ANN10 + T/P DeepONet10	0.2	7.87e-03	0.0701	0.265	0.595	0.333	0.238	0.0622	0.996	0.999	0	18/18
5 ANN10 + T/P DeepONet10	0.3	0.0117	0.0733	0.286	0.646	0.365	0.196	0.103	1	1	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.16	0.499	1.42	3.36	0.699	1.4	0.313	6.87	5.37	0	18/18
5 ANN10 + T/P DeepONet10	1	0.242	0.721	2.28	2.68	0.749	0.599	0.285	1.08	16.6	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Test T/P/ϕ DeepONet10: common ICs [4, 5, 7, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	0.0966	0.155	1	0.991	0.256	1	0.299	1.01	1	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.0835	0.0903	1	1	0.345	1	0.134	1.41	1.92	0	17/18
6 T/P/ϕ DeepONet10	0.2	0.0148	0.115	0.583	0.73	0.391	1	0.101	0.998	1	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.0883	0.45	1.95	1	0.558	1	0.304	0.999	1	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.229	0.552	8.87	5.14	0.618	3.39	0.288	4.87	2	0	18/18
6 T/P/ϕ DeepONet10	1	0.29	0.847	2.2	2.69	0.837	0.706	0.355	1.09	8.21	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Test ANN10 + T/P/ϕ DeepONet10: common ICs [4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0836	0.331	0.998	0.995	0.489	1.72	0.382	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0633	0.0986	1	0.584	0.195	0.998	0.192	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0124	0.124	0.476	0.512	0.378	0.288	0.0502	0.997	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0468	0.334	1	0.783	0.401	0.943	0.0996	0.994	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.122	0.618	3.29	1.83	0.622	1.11	0.208	1.39	0.997	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.206	0.646	2.83	1.65	0.759	0.441	0.219	1.35	13.8	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.